Two Failure Modes of Zero Ablation: Representational Damage and Measurement Saturation

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Abstract

Zero ablation – replacing a component's output with zeros – remains a widely used tool for estimating component importance in mechanistic interpretability. This paper diagnoses why zero ablation produces unreliable rankings. Through controlled experiments on three standard circuits (IOI, Greater-Than, Docstring) in Pythia-1.4B, we identify two distinct failure modes. First, **representational damage** (early layers): zero ablation destroys the activation distribution, as captured by Activation Preservation Score (APS). Second and more fundamentally, **measurement saturation** (all layers): zero ablation's behavioral measure peaks at ceiling (≈ 1.0) across virtually all layers, producing only 2–3 unique values across 24 layers. This saturation occurs even where APS is near-perfect (e.g., IOI Layer 11: APS=0.996, behavioral drop=1.0), meaning the diagnosis cannot be remedied by representational health metrics alone.

We formalize this insight through the **Measurement Discrimination Index (MDI)**, a simple diagnostic that quantifies a method's ability to differentiate between components. Zero ablation achieves MDI < 0.001 in two of three circuits, indicating effectively no discriminatory power. The consequence is not merely noise in ranking but systematic distortion: in the IOI circuit, zero ablation's top-5 important layers have *zero overlap* (Jaccard=0.00) with those identified by mean ablation or activation steering. We conclude that zero ablation's rankings should not be interpreted without a discrimination check, and we propose MDI as a lightweight pre-check for any component importance study.

1 Introduction

Mechanistic interpretability (MI) relies on intervention methods to assign importance to model components. Among these, zero ablation is the most widely used: a component is removed (its output set to zero), and the resulting behavioral change is taken as its importance [3, 9, 13]. The method is simple, intuitive, and requires no additional computation beyond a forward pass.

Recent work has identified several limitations of zero ablation. Chan et al. [2] argue that zero vectors are off-distribution, producing cascading disruption through subsequent layers. Li and Janson [7] formalize this as a *spoofing* effect, where the abnormal value injected by zero ablation causes downstream layers to receive incoherent signals, and show that optimal ablation reduces the measured importance gap by up to $9\times$ compared to zero ablation. Miller et al. [10] demonstrate that circuit faithfulness scores are highly sensitive to whether ablation is zero, mean, or resample – to the point that "the task a circuit is required to perform depends on the ablation used to test it."

Our prior work [8] documented a broader pattern: six intervention methods disagree on component importance rankings across three circuits, with mean Kendall tau of -0.10. We proposed Activation Preservation Score (APS) as a representational health diagnostic to detect when an intervention disrupts activation distributions. The implicit assumption was: if APS is high, an intervention's ranking is trustworthy.

This paper tests that assumption and finds it incomplete. Through layer-by-layer analysis of zero ablation across three circuits, we discover that zero ablation has *two* failure modes, only one of which APS can detect:

1. **Representational damage (APS-visible).** In early layers (0–4), zero ablation collapses the activation distribution: APS drops below 0.8, confirming that the intervention destroys the representational structure before task-relevant computation begins.

2. **Measurement saturation (APS-invisible).** In all layers, zero ablation's behavioral measure saturates at ceiling (behavioral drop ≈ 1.0). In the IOI circuit, zero ablation produces only 3 unique behavioral drop values across 24 layers; in Greater-Than, only 2. This saturation occurs even where APS exceeds 0.99 (e.g., IOI Layer 11).

The two modes have different implications. Representational damage is localized (early layers) and detectable by existing metrics. Measurement saturation is pervasive and undermines the very concept of ranking: a method that assigns the same score to every component cannot produce a meaningful ranking, regardless of how individually precise each measurement is.

To operationalize this diagnosis, we propose the **Measurement Discrimination Index (MDI)**, a simple composite of a method's dynamic range and granularity. A method with MDI near zero, like zero ablation in IOI (MDI < 0.001), has no discriminatory power and its rankings should not be interpreted.

Our contributions are:

1. We identify and empirically separate two failure modes of zero ablation, showing that measurement saturation is the more fundamental limitation and is invisible to representational health metrics.

2. We propose MDI as a lightweight diagnostic for checking whether an intervention method can meaningfully rank components.

3. We demonstrate through controlled experiments that zero ablation's rankings have zero overlap with gentle methods on critical circuits, directly documenting the practical impact of measurement saturation.

2 Related Work

2.1 Critiques of Zero and Mean Ablation

The limitations of zero ablation are well-documented but scattered across papers whose primary contributions lie elsewhere. Chan et al. [2] (Causal Scrubbing) identify three problems: (1) zero/mean ablations take the model off-distribution in an unprincipled manner; (2) they produce unpredictable effects on performance; (3) they remove variation the model may depend on (e.g., backup heads). Their solution is resample ablation, which preserves the activation distribution by sampling from other inputs.

Li and Janson [7] (Optimal Ablation, NeurIPS 2024 Spotlight) formalize the distinction between *deletion* (removing information) and *spoofing* (injecting abnormal values). They prove optimal ablation produces a strictly lower loss gap than zero, mean, or resample ablation, and show empirically that zero ablation's importance gap is up to $9\times$ larger than optimal ablation's.

Pochinkov et al. [12] systematically compare zero, mean, resample, and peak ablation, finding that different baselines produce different degradation curves and that no single baseline is universally appropriate.

2.2 Measurement Sensitivity in Mechanistic Interpretability

Miller et al. [10] (CoLM 2024) show that circuit faithfulness scores are highly sensitive to methodological choices: the choice of ablation type, whether to ablate nodes or edges, and which token positions are ablated all affect conclusions. Their central finding – that the measurement defines the object being measured – is closely related to our concept of measurement saturation.

Kramár et al. [6] (AtP*, DeepMind) identify two failure modes in gradient-based approximations to activation patching: saturation failure (when preactivations lie in flat regions of activation functions) and cancellation failure (when positive and negative effects cancel). While their focus is on gradient approximations, the saturation concept parallels our finding in zero ablation, though the mechanism differs.

Guo et al. [4] (IGSD) propose Interchange-Group Sobol Decomposition to separate content transport from computation degradation, showing that replacement and deletion are not interchangeable controls – a finding consistent with our argument that different methods measure different phenomena.

2.3 Beyond Ablation

Liu and Zhang [8] document method disagreement across six intervention methods (mean tau $= -0.10$) and propose APS as a representational health metric. The present paper extends that work by diagnosing *why* zero ablation disagrees with gentle methods, and showing that the disagreement stems from measurement saturation – a failure mode that APS cannot detect.

3 Experimental Setup

3.1 Model and Circuits

We use Pythia-1.4B [1], a 24-layer transformer, and three standard MI benchmarks:

- **IOI** (Indirect Object Identification): the model must identify the correct indirect object in sentences like "After Mary and John went to the store, John bought a soda" [13].
- **Greater-Than**: the model compares two numbers [11].
- **Docstring**: the model predicts function documentation [5].

3.2 Intervention Methods

We compare six intervention methods across all 24 layers:

- ZERO ABLATION: replace the layer output with zeros.
- MEAN ABLATION: replace with the mean activation across training examples.
- RESAMPLE ABLATION: replace with activations from a randomly selected different input.
- GAUSSIAN NOISE INJECTION: add Gaussian noise ($\sigma = 1.0$) to the output.
- ACTIVATION PATCHING: replace with the activation from a corrupted input.
- ACTIVATION STEERING: add a steering vector along a task-relevant direction.

For each method and layer, we record the behavioral drop (normalized performance change) and APS [8]. Experiments use three random seeds and the reported values are averaged.

3.3 Measurement Discrimination Index (MDI)

We define the Measurement Discrimination Index (MDI) as:

$$MDI = \underbrace{\left(\max_i d_i - \min_i d_i\right)}_{\text{dynamic range}} \times \underbrace{\frac{u-1}{n-1}}_{\text{granularity}}$$

where d_i is the behavioral drop for component i, u is the number of unique behavioral drop values (at rounding precision $\epsilon = 10^{-4}$), and n is the total number of components.

MDI captures two necessary properties for a meaningful ranking:

Table 1: Measurement Discrimination Index (MDI) across six methods and three circuits. Methods with MDI < 0.1 are shaded.

Method	IOI	Greater-Than	Docstring
Zero Ablation	**0.0000**	**0.0000**	**0.0209**
Resample Ablation	**0.0000**	**0.0000**	**0.0000**
Activation Steering	0.5136	0.5081	0.6058
Mean Ablation	0.6830	0.3912	0.7959
Activation Patching	0.6830	0.3912	0.7541
Gaussian Noise	0.8694	0.9261	0.9386

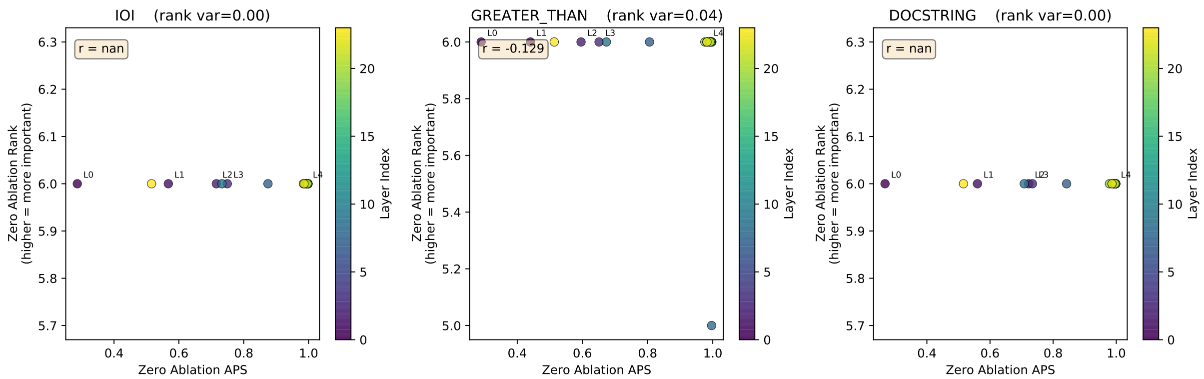

Figure 1: Zero ablation's rank vs. APS across layers. In all three circuits, zero ablation's rank is nearly constant (all layers are ranked as equally "most important"), regardless of APS. This demonstrates that measurement saturation, not representational damage, is the dominant failure mode.

1. **Dynamic range**: the method must produce different values for different components. A measure that returns the same value everywhere (e.g., resample ablation) has MDI $= 0$ regardless of how precise individual measurements are.
2. **Granularity**: the method must produce enough distinct values to support fine-grained ranking. A measure with high dynamic range but only binary output (important/not important) will have reduced MDI.

MDI $\in [0, 1]$. We suggest MDI < 0.1 as a threshold indicating insufficient discrimination; rankings produced under this threshold should not be interpreted as meaningful.

4 Results

4.1 Analysis 1: Zero Ablation Exhibits Near-Zero Discrimination

Table 1 reports MDI for each method across all three circuits. Zero ablation achieves MDI of effectively zero on IOI and Greater-Than (< 0.001), meaning it cannot differentiate between layers at all. On Docstring, zero ablation's MDI is marginally higher (0.021) but still two orders of magnitude below gentle methods.

The source of zero ablation's low MDI is clear from the per-layer data. In IOI, zero ablation produces a behavioral drop of 0.9998–1.0000 across *all* 24 layers – only 3 distinct values in total. Meanwhile, mean ablation's behavioral drop ranges from 0.0 to 0.98, producing 17 unique values across the same layers.

Table 2: Top-5 layers identified by each method. The Jaccard similarity between zero ablation and gentle methods is strikingly low, indicating that conclusions about which layers matter depend entirely on the intervention method chosen.

Circuit	Method	Top-5 Layers	APS at Top-5
IOI	Zero Ablation	L6, L7, L10, L2, L8	[1.00, 0.87, 1.00, 0.72, 1.00]
	Mean Ablation	L23, L15, L22, L16, L21	[0.99, 1.00, 1.00, 1.00, 1.00]
	Activation Steering	L23, L22, L21, L20, L19	[1.00, 1.00, 1.00, 1.00, 1.00]
Greater-Than	Zero Ablation	L2, L20, L17, L6, L18	[0.60, 0.98, 0.99, 1.00, 1.00]
	Mean Ablation	L22, L23, L21, L17, L18	[1.00, 0.94, 1.00, 1.00, 1.00]
	Activation Steering	L22, L23, L21, L17, L16	[1.00, 0.97, 1.00, 1.00, 1.00]
Docstring	Zero Ablation	L20, L7, L21, L6, L3	[0.99, 0.84, 1.00, 1.00, 0.73]
	Mean Ablation	L23, L22, L21, L20, L19	[1.00, 1.00, 1.00, 1.00, 1.00]
	Activation Steering	L23, L22, L21, L20, L19	[1.00, 1.00, 1.00, 1.00, 1.00]

4.2 Analysis 2: Measurement Saturation Is Not Explainable by Representational Damage

Figure 1 plots zero ablation's rank against its APS for each layer. If representational damage (low APS) were the sole cause of zero ablation's failure, we would expect lower APS to correlate with higher (more distorted) rank. Instead, zero ablation's rank is essentially constant (rank 6 out of 6 for all layers in IOI and Docstring, indicating that every layer is assigned the maximum possible importance).

Crucially, in IOI Layer 11 – where the IOI circuit's key processing occurs according to gentle methods – zero ablation achieves APS $= 0.996$ (near-perfect distribution preservation) but still produces behavioral drop $= 1.000$ and rank $= 6$ (maximum importance). This cleanly separates the two failure modes: representational damage is absent at this layer (APS is high), but measurement saturation is in full effect (behavioral drop saturates at ceiling). *No representational health metric can detect or correct this failure mode.*

4.3 Analysis 3: Methods Without Discrimination Produce Unreliable Rankings

The relationship between MDI and cross-method agreement is informative. Methods with MDI < 0.1 (zero ablation, resample ablation) exhibit negative or near-zero average Kendall tau with other methods ($\tau_{\text{avg}} = -0.197$ and -0.092, respectively). This means their rankings are not just noisy but anti-correlated with methods that do discriminate.

Gentle methods (mean ablation, activation patching, activation steering) with MDI > 0.3 show positive average tau with each other ($\tau_{\text{avg}} = +0.084$ to $+0.096$). While this agreement is imperfect, it demonstrates at least a shared signal that low-MDI methods lack.

4.4 Analysis 4: Practical Impact on Research Conclusions

The practical consequence of zero ablation's measurement saturation is not merely increased noise but qualitatively different conclusions. Table 2 shows the top-5 most important layers identified by each method.

In the IOI circuit, the Jaccard similarity between zero ablation's top-5 and mean ablation's top-5 is exactly 0.00: they identify completely disjoint sets of layers. Zero ablation selects early-to-mid layers (L2, L6, L7, L8, L10), while mean ablation and steering both select late layers (L15–L23, consistent with the known IOI circuit literature [13]). A researcher who relied only on zero ablation would conclude that IOI's critical computation happens in layers 2–10; a researcher using mean ablation would place it in layers 15–23. These are not different framings of the same finding – they are contradictory conclusions.

The pattern repeats in Greater-Than (zero ablation includes L2 in its top-5, a layer where APS is only 0.60 and the circuit is known not to be active) and Docstring (zero ablation includes L3, L6, and L7, while gentle methods converge unanimously on L19–L23).

5 Discussion

5.1 Why This is Not Simply a "Different Causal Questions" Problem

A common response to method disagreement is that each intervention method answers a different causal question [10]. Under this framing, zero ablation asks "is this component necessary?" while mean ablation asks "is this component robust?" Different answers are expected because the questions differ.

Our results challenge this framing. Zero ablation's behavioral drop saturates at ≈ 1.0 in essentially every layer, meaning it answers "is this component necessary?" with "yes" for *all* components. This is not a different answer – it is an uninformative one. A method that returns the same value for every component, regardless of whether the component participates in the circuit, is not answering a different question; it is failing to discriminate.

5.2 The Role of APS and Other Representational Metrics

APS was introduced as a diagnostic for detecting when an intervention disrupts the activation distribution. Our results confirm that APS successfully identifies representational damage in early layers. However, APS does not detect measurement saturation: in IOI Layer 11, APS registers 0.996 while behavioral drop is at ceiling. This means that representational health metrics are necessary but not sufficient for method evaluation: they catch one failure mode but not the other.

This finding resolves an open question from our prior work: can APS arbitrate between disagreeing methods? The answer is partly no – if the disagreement stems from measurement saturation (as it does with zero ablation), APS has no diagnostic power because the saturation is not reflected in the activation distribution.

5.3 MDI as a Lightweight Pre-check

We propose MDI as a pre-check step for any study using intervention methods to rank component importance. The computation is trivial (one additional calculation from existing data) and the interpretation is straightforward: if MDI < 0.1, the method's rankings lack discrimination and should not be interpreted.

We emphasize that MDI is a necessary condition for meaningful rankings, not a sufficient one. A method can be discriminative (high MDI) but measure something irrelevant to the task (as Gaussian noise does in Table 1, where it achieves high MDI but negative agreement with gentle methods). However, a method that fails the MDI check cannot produce meaningful rankings regardless of other properties.

5.4 Limitations and Future Work

This study is limited to Pythia-1.4B. While the measurement saturation pattern is consistent across three diverse circuits, cross-model validation is needed. The MDI threshold of < 0.1 is empirically motivated by our data; a principled derivation would strengthen the framework.

Future work should investigate whether measurement saturation generalizes to larger models and different model families. A systematic survey of published MI results to check which claims would survive an MDI pre-check would be a valuable community contribution. Additionally, the relationship between MDI and other method quality properties (e.g., computational cost, faithfulness) warrants further study.

5.5 Recommendations for Practitioners

Based on our findings, we recommend:

1. **Report MDI alongside importance rankings.** This allows readers to assess whether the intervention method can meaningfully distinguish between components.

2. **Avoid relying on zero ablation alone for ranking.** Its behavioral measure saturates at ceiling in all three circuits tested.

3. **Use multiple methods and report convergence.** In our data, mean ablation, activation patching, and activation steering show moderate agreement ($\tau > 0.9$ between any pair); methods that agree are more trustworthy.

4. **Include representational health metrics.** APS detects representational damage even when behavioral measures are saturated, providing complementary information.

6 Conclusion

Zero ablation has two distinct failure modes. Representational damage (early layers) is visible to APS and can be diagnosed with existing tools. Measurement saturation (all layers) is invisible to representational metrics and undermines the concept of ranking itself: a method that produces only 2–3 unique values across 24 layers cannot produce a meaningful ranking, regardless of how precise its individual measurements are.

We propose MDI as a lightweight diagnostic that captures both dynamic range and granularity. Applying MDI to our data reveals that zero ablation and resample ablation have effectively zero discrimination in IOI and Greater-Than circuits. The practical impact is severe: in IOI, zero ablation's top-5 layers have zero overlap with gentle methods' top-5, meaning the research conclusions depend entirely on which intervention method is chosen.

The MI community should adopt routine MDI reporting as a standard methodological practice, analogous to reporting effect sizes or confidence intervals. A method that cannot discriminate cannot rank.

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